

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year EVEN Semester Examination 2024

COURSE NO: Phy 1213 COURSE TITLE: Physics

FULL MARKS: 60

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks and Course Outcomes (COs).
(iii) Use separate answer script for each section.

SECTION : A		Marks	COs
Q.1.	(a) Define Miller indices of the crystal planes. Obtain a relation between interplaner distance and Miller indices.	03	CO1
	(b) Deduce Bragg's law of x-ray diffraction through crystals. Why only first order diffraction is considered while applying this law to determine crystal structure.	03	CO1
	(c) Calculate packing fraction for bcc and fcc structure.	02	CO3
	(d) Explain thermal excitation in semiconductor.	02	CO1
✓Q.2.	(a) Explain the term coherent sources. How are they realized in practice? Discuss how coherent sources are produced in Fresnel's bi-prism.	03	CO2
	(b) Explain how Newton's ring are formed. How would you employ this phenomenon for measuring the wavelength of light?	04	CO2
	(c) In a Newton's rings experiment the diameters of the 5 th and 15 th bright rings are 0.336 cm and 0.590 cm respectively. If the radius of the Plano-convex lens is 100 cm. Calculate the wavelength of light used.	03	CO3
Q.3.	(a) Derive the intensity distribution for Fraunhofer diffraction from a single slit and deduce the condition for maxima and secondary minima.	05	CO2
	(b) Derive the missing orders in a double slit diffraction pattern.	03	CO2
	(c) Yellow doublet of Na consists of 5890Å and 5896Å. Calculate the minimum number of lines that a grating must have so as to resolve this doublet in the third order.	02	CO3
Q.4.	(a) State Brewster's law. Verify from this law that when light is incident on the transparent substance at the polarizing angle, the reflected and refracted rays are at right angles.	03	CO2
	(b) Discuss the theory of double refraction in a calcite crystal with necessary diagram. Distinguish between ordinary and extra-ordinary rays.	04	CO2
	(c) How will you orient the polarizer and the analyzer so that a beam of natural light is reduced to 0.25 of its original intensity?	03	CO3
SECTION : B			
✓Q.5.	(a) Derive the differential equation of the motion of the mass in the spring-mass system. Describe the nature of the motion by solving the differential equation of motion.	04	CO2
	(b) Explain the effect of heavy and less damping on the displacement of a damped harmonic oscillator with appropriate examples.	03	CO2
	(c) Find the resultant motion of two perpendicular simple harmonic vibrations of same frequency but different phases and amplitudes. Draw the Lissajous figures for $\phi=\pi/2$ and $\phi=\pi$.	03	CO3
Q.6.	(a) Analytically discuss the formation of nodes and antinodes and hence show that two nodes or two antinodes are separated from each other by a distance $\lambda/2$.	04	CO2
	(b) Obtain an expression for a plane progressive wave travelling in the positive x-direction in the form $y = a \sin \frac{2\pi}{\lambda}(vt - x)$, where the symbols have their usual meanings.	03	CO2
	(c) Show that if the source moves away with the velocity of sound from an observer who is at rest, the frequency of the vibration is halved.	03	CO3
Q.7.	(a) Show the experimental results of photo-electric effect. Explain the failure of classical electromagnetic theory to explain the results of photo-electric effect.	03	CO1
	(b) Obtain the expression for the change in wavelength of the X-ray photon in Compton scattering.	04	CO1
	(c) Use the results of Robert Millikan's experiment to verify Einstein's photo-electric equation.	03	CO3
Q.8.	(a) Based on Bohr's assumption, derive an expression for the orbital energy of an electron in the hydrogen atom.	04	CO1
	(b) Show that the group velocity is given by the relation $v_g = v_p - \lambda \frac{\Delta v_p}{\Delta \lambda}$, where the symbols have their usual meanings.	04	CO1
	(c) Calculate the de-Broglie wavelength of a beam of electron whose energy is 100 eV.	02	CO3

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FULL MARKS: 60

TIME: 3 HOURS

N.B.

- (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.
(iv) Course Outcomes (COs)

<u>SECTION : A</u>			Marks	COs
Q.1.	(a)	Sketch the (0 2 0) and (1 $\bar{1}$ 1) plane in a simple cubic crystal.	02	CO1
	(b)	Define packing fraction. Determine the value of packing fraction for body-centered cubic and face-centered cubic structure.	05	CO1
	(c)	Discuss the various types of bondings in crystals giving examples.	03	CO1
Q.2.	(a)	Compare the outcomes of wave theory and quantum theory of light.	02	CO2
	(b)	Show that in the Young's double slit experiment, the resultant intensity at a point has the $\cos^2 \phi/2$ dependence where ϕ is the phase difference between the waves arriving at that point.	04	CO2
	(c)	Explain the formation of Newton's rings. How are Newton's rings used for determining the refractive index of a liquid?	04	CO2
Q.3.	(a)	In Fraunhofer diffraction due to a single slit, show that the intensity of the first secondary maxima is roughly 4.5% of that of the principal maximum.	04	CO2
	(b)	Sketch and explain the intensity distribution curves for single slit, double slit and multiple slit of Fraunhofer diffraction pattern.	03	CO2
	(c)	Light of wavelength 5000Å is incident normally on a plane diffraction grating having 14000 lines per inch. How can many orders of diffracted images be observed?	03	CO3
Q.4.	(a)	Explain the term polarization of light. How would you obtain plane polarized light by reflection?	03	CO2
	(b)	State and explain the law of Malus to obtain the intensity of plane polarized light.	03	CO2
	(c)	Describe the construction of a Nicol Prism and explain how it produces plane polarized light.	04	CO2
<u>SECTION : B</u>				
Q.5.	(a)	Derive the differential equation of the motion of the charge in an LC circuit and hence show that the current in the circuit is oscillatory in nature.	03	CO2
	(b)	Find the resultant motion of two perpendicular simple harmonic vibrations of periods in the ratio 1:2 and having (i) Zero phase difference and (ii) phase difference of $\pi/2$.	04	CO3
	(c)	Define the quality factor of a damped oscillator. Show that the less the damping the better will be the quality of the oscillator.	03	CO2
Q.6.	(a)	Derive Einstein's photo-electric equation and show that the kinetic energy of the photo-electrons depends upon the frequency of the incident radiation and not on its intensity.	04	CO1
	(b)	Explain the Compton effect. Why is the Compton shift not observed for visible light?	03	CO1
	(c)	Light having a wavelength of 5000Å falls on a material having work function of 1.90eV. Calculate (i) the maximum kinetic energy of the ejected photo-electrons and (ii) the stopping potential.	03	CO3
Q.7.	(a)	Obtain an expression for a plane progressive wave travelling in the positive x-direction.	03	CO2
	(b)	Analytically discuss the formation of beats and show that the number of beats per second is equal to the difference in frequency of the two notes.	04	CO2
	(c)	A source of sound has an amplitude of 0.25 cm and a frequency of 512 Hz. If the velocity of sound in air is 340 m/s and the density of air is 0.0012 g/cm ³ then find the rate of flow of energy per square cm.	03	CO3
Q.8.	(a)	Explain the origin of spectral lines on the basis of Bohr atomic model.	03	CO1
	(b)	Show that the de Broglie groups velocity is equal to the velocity of the particle.	02	CO1
	(c)	Apply the results of Davisson and Germer experiment to verify matter wave by observing electron beam diffraction through crystals.	03	CO1
	(d)	10kV electrons passing through a thin film of metal with atomic spacing $d = 5.6 \times 10^{-10}$ m. Calculate the angle of deviation of the first order diffraction maximum.	02	CO3

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2022

COURSE NO: Phy 1213

COURSE TITLE: Physics

FULL MARKS: 60

TIME: 3 Hours

N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.

(ii) Figures in the right margin indicate full marks.

(iii) Use separate answer script for each section.

(iv) COs- Course Outcomes

SECTION A

COs Marks

- | | | | |
|----------|--|-----|----|
| Q.1. (a) | The atoms are closely packed in face centered cubic structure than body centered cubic structure. Justify your answer. | CO1 | 03 |
| (b) | Based on Bohr's assumption, obtain an expression for the orbital energy of an electron in the hydrogen atom. | CO1 | 04 |
| (c) | Calculate the radius of the first Bohr's orbit for <u>He</u> atom and also the velocity of the electron in these orbit. | CO3 | 03 |
| Q.2. (a) | Mention the drawbacks of Huygens's principle of secondary waves and explain how they are overcome by the succeeding theories. | CO2 | 04 |
| (b) | Show that the interference of two light waves causes the redistribution of light energy and the resultant pattern is in accord with the law of conservation of energy. | CO2 | 04 |
| (c) | Mention few applications of interference in thin films. | CO2 | 02 |
| Q.3. (a) | When does the diffraction of waves become noticeable? Distinguish between the Fresnel and Fraunhofer classes diffraction. | CO2 | 03 |
| (b) | Give the construction and theory of plane transmission grating and show how you would use it to find the wavelength of light. | CO2 | 04 |
| (c) | Deduce the missing orders for a double slit Fraunhofer diffraction pattern if the slit widths are 0.16mm and they are 0.80mm apart. | CO3 | 03 |
| Q.4. (a) | Explain the action of tourmaline crystal and pile of plates in producing polarization. | CO2 | 04 |
| (b) | What is meant by optical activity of a substance? Distinguish between dextro-rotary and leavo-rotary substances. | CO2 | 03 |
| (c) | How is half-shade polarimeter used to find the specific rotation of a substance? | CO2 | 03 |

SECTION B

- | | | | |
|----------|--|-----|----|
| Q.5. (a) | Establish Einstein's photo-electric equation. Show that the stopping potential varies linearly with the frequency of the incident radiation but is independent of its intensity. | CO1 | 04 |
| (b) | Describe the construction and working of a photo-voltaic cell. | CO1 | 03 |
| (c) | An X-ray photon having wavelength of $0.3\text{\AA}$ collides with an electron at rest and is scattered through 60° . Calculate the wavelength of the scattered photon. | CO3 | 03 |
| Q.6. (a) | Explain Doppler's principle in sound. Obtain an expression for the apparent frequency of a note when the source and the observer are moving towards each other. | CO2 | 03 |
| (b) | Show that the group velocity is given by the relation, $V_g = V_p - \lambda \frac{dV_p}{d\lambda}$, where the symbols have their usual meaning. | CO1 | 04 |
| (c) | A bus approaches a rest observer at a speed of 60 kilometers per hour sounding a horn of frequency 750Hz. Determine the apparent frequency of the horn to the observer. (Velocity of sound = 340 m/sec.) | CO3 | 03 |
| Q.7. (a) | Show that the principle of superposition is valid only in case of linear vibrations. | CO2 | 04 |
| (b) | Discuss the conditions under which the oscillations become oscillatory in damped oscillator. | CO2 | 04 |
| (c) | Write down the differential equation for damped and forced oscillator. | CO2 | 02 |
| Q.8. (a) | Differentiate stationary waves from progressive. | CO2 | 03 |
| (b) | For a plane progressive wave, show that on the average, half the energy is kinetic and half potential. | CO2 | 04 |
| (c) | A wave of frequency 400Hz is travelling with a velocity 800 m/s. How far two points are situated whose displacements differ in phase by $\pi/4$. | CO3 | 03 |

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2021

COURSE NO: Phy 1213

COURSE TITLE: Physics

FULL MARKS: 60

TIME: 3 Hours

- N.B. (i) Answer any **SIX** questions taking any **THREE** from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

Marks

SECTION A

- Q.1. (a) What are lattice, basis and Bravais lattices? [CO1] 03
(b) On the basis of energy band structure, how are the crystalline solids classified into conductors, semiconductors and insulators? [CO1] 04
(c) Show that the lattice constant for a cubic cell is given by $a = \sqrt[3]{nm/\rho N}$. [CO1] 03
- Q.2. (a) Explain how Bohr and Einstein contributed to the quantum mechanics. [CO2] 03
(b) Describe the complementary nature of fringes with reflected and transmitted light in thin films. [CO2] 03
(c) How many orders will be visible if the wavelength of light is $5000 \text{ \AA}$ and the number of lines per inch on the grating is 2620? [CO3] 04
- Q.3. (a) What is Newton's ring and how are they formed? [CO2] 03
(b) Sketch the intensity distribution curves for single, double and multiple slits of Fraunhofer diffraction pattern. [CO2] 03
(c) The diameters of the n -th and $(n + 8)$ -th bright rings are 4.2 mm and 7.0 mm respectively in a Newton's rings system. If the radius of curvature of the lower surface of the lens is 2 m, what will be the wavelength of light and the ring number, n ? [CO3] 04
- Q.4. (a) What is meant by the term polarization of light? Draw polarized, partially polarized and unpolarized light in physical optics notation. [CO2] 03
(b) State Brewster's law. Show that at the polarizing angle of incidence the reflected and refracted rays are mutually perpendicular to each other. [CO2] 03
(c) Discuss the construction of a Nicol prism and explain how it produces plane polarized light. [CO2] 04

SECTION B

- Q.5. (a) State Einstein photo-electric equation. Discuss how this equation provides a satisfactory explanation of the observed facts regarding photo-electric emission. [CO1] 03
(b) Analytically discuss the formation of beats and show that the number of beats per second is equal to the difference in frequency of the two notes. [CO2] 04
(c) A surface having work function 1.51 eV is illuminated by light of wavelength $4000 \text{ \AA}$. Calculate (i) the maximum kinetic energy of the ejected electrons and (ii) the stopping potential. [CO3] 03
- Q.6. (a) Explain Compton effect. Why is the Compton shift not observed for visible light? [CO1] 04
(b) Show that the velocity of matter wave is greater than the velocity of light. [CO1] 03
(c) Calculate the de-Broglie wavelength associated with electrons whose speed is 2% of the speed of light. [CO3] 03
- Q.7. (a) Show that the particle executing simple harmonic motion obeys the law of conservation of energy. [CO2] 04
(b) Discuss the formation of Lissajous' figure having the same time periods of two simple harmonic motions. [CO2] 04
(c) A body describing simple harmonic motion has a maximum acceleration of $8\pi \text{ m/s}^2$. Find the time period. [CO3] 02
- Q.8. (a) Distinguish between particle velocity and wave velocity and obtain a relation between the two. [CO2] 03
(b) Show that the energy of a plane progressive wave is given by $E = 2\pi^2 a^2 n^2 \rho$; where the symbols have their usual meaning. [CO2] 04
(c) A source of sound has an amplitude and frequency of 0.25 cm and 512 Hz respectively. If the velocity of sound in air is 250 m/s and the density of air is 0.00129 gm/cm^3 , what is the rate of flow of energy per square cm? [CO3] 03

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
1st Year Even Semester Examination 2019
COURSE NO: Phy 1213 COURSE TITLE: Physics
FULL MARKS: 72 TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
(ii) Figures in the right margin indicate full marks.
(iii) Use separate answer script for each section.

SECTION : A**Marks**

- Q.1. (a) What were the main results of Rutherford's alpha scattering experiment? How these results ruled out Thomson's atom model. 3
(b) Obtain a general expression for (i) the radius of Bohr orbit and (ii) the orbital frequency of electron occupying them. 6
(c) The ionization energy of H-like atom is 4 Rydberg. What is the wavelength of radiation emitted when the electron jumps from the first excited state to the ground state? $1 \text{ Rydberg} = 2.2 \times 10^{-10} \text{ J}$. 3
- Q.2. (a) Draw a curve showing stopping potential against frequency of a photo sensitive material. How do you determine Plank's constant with the help of the graph? 3
(b) A beam of monochromatic X-rays of wavelength λ is scattered by a light element. Obtain an expression for the wavelength of scattered radiation. 6
(c) Photon of energy 1.02 MeV undergoes Compton scattering through 180° . Calculate the energy of scattered photon. 3
- Q.3. (a) Define simple harmonic motion. Show that simple harmonic motion can be regarded as a projection of uniform circular motion on the diameter of a circle. 4
(b) What are Lissajous' figures? Discuss the formation of Lissajous' figures when the periods of the two vibrations are equal and the phase difference is 0 and $\pi/2$. 6
(c) A body is vibrating with simple harmonic motion of amplitude 13 cm and frequency 4 Hz. Compute the maximum values of the acceleration and velocity. 2
- Q.4. (a) Distinguish between phase velocity and group velocity and obtain a relation between the two. 5
(b) Show that for a plane progressive wave on the average half the energy is kinetic and half potential. 5
(c) A train approaches a stationary observer at a speed of 80 kilometers per hour sounding a whistle of frequency 1000 Hz. What will be the apparent frequency of the whistle to the observer? (velocity of sound = 332 m/s) 2

SECTION : B

- Q.5. (a) What is crystal? What are the lattice parameters of a unit cell? 2
(b) Calculate the effective number of lattice points of simple cubic and face centered cubic structure of crystal. 2
(c) Describe the seven systems of crystals along with their characteristics. 4
(d) Calculate the value of packing fraction for a b.c.c and f.c.c crystal structure. 4
- Q.6. (a) Explain the formation of a stable bond using the potential energy versus inter atomic distance curve. 5
(b) Explain the various types of bonding in crystals. Illustrate with example. 4
(c) Distinguish between a metal, semiconductor and insulator according to band theory. 3
- Q.7. (a) Explain with necessary diagram the theory of interference due to transmitted light. In this case why are the fringes less distinct? 5
(b) What do you understand by resolving power and dispersive power of grating? Prove that the dispersive power of the grating is directly proportional to the order of the spectrum. 4
(c) Light of wavelength λ is incident normally on a plane diffraction grating having N lines per unit length. How many orders of diffracted images can be observed? 3
- Q.8. (a) What is meant by plane polarized light, circularly polarized light and elliptically polarized light. How are they produced and detected? 6
(b) What is a quarter-wave plate? Deduce its thickness for a given wavelength in terms of its refractive indices. 4
(c) A 20 cm long tube containing sugar solution rotates the plane of polarization by 11° . If the specific rotation of sugar is 66.5° , calculate the strength of the solution. 2

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

1st Year Even Semester Examination 2018

COURSE NO: Phy 1213 COURSE TITLE: Physics

FULL MARKS: 72

TIME: 3 HRS

- N.B. (i) Answer any SIX questions taking any THREE from each section.
 (ii) Figures in the right margin indicate full marks.
 (iii) Use separate answer script for each section.

SECTION : A

- Q.1. (a) Give an account of Rutherford's theory of scattering of α -particles and describe how it has been verified. Mention the deficiencies in this model. 6
 (b) Derive an expression for the Rydberg constants. Show how the limits of Lyman and Balmer series are related to each other. 4
 (c) Calculate the frequency of the electron in the Bohr's first orbit in a hydrogen atom. 2
- Q.2. (a) Define photoelectrons and photoelectric work function. Establish Einstein's photo-electric equation. 4
 (b) Write down the properties of matter waves. Obtain an expression for the De-Broglie wavelength of matter waves. 5
 (c) An X-ray photon collides with an electron at rest and is scattered through 90° . What is its frequency after collision? Its initial frequency is 3×10^{19} Hz. 3
- Q.3. (a) Give examples of the systems that oscillate with one degree of freedom. Find an expression for the frequency of oscillations of an LC circuit. 4
 (b) Under what condition the oscillations of a damped harmonic oscillator become oscillatory? 4
 (c) What is called the quality factor of a damped oscillator? If the quality factor of a sonometer wire of frequency 300 cps be 2×10^3 , in what time will its energy be reduced to $\frac{1}{e}$ th of its energy in the absence of damping? 4
- Q.4. (a) Find the differential equation of a one dimensional plane progressive wave. Which of the following are the solutions of plane progressive wave? (i) $y = (x - vt)^2$ and (ii) $y = \sin 2x \cos vt$. 5
 (b) Distinguish between interference of sound and beat. Show that the number of beats produced per second is equal to the difference in frequency of the two notes. 4
 (c) When a train is approaching the observer, the frequency of the whistle is 100 cps while when it passed the observer it is 50 cps. Calculate the frequency when the observer moves with the train. 3

SECTION : B

- Q.5. (a) What are Miller indices? Why are they important for solid? 3
 (b) Draw a schematic diagram of the unit cell for the SC, BCC and FCC crystal structures. 3
 (c) Draw (101), (111) and (001) planes in a cubic unit cell. 3
 (d) Write a short note on "Braggs law of X-ray diffraction". 3
- Q.6. (a) How can coherent sources be obtained in practice? 3
 (b) Give the theory of Newton's ring in thin film interference. How would you obtain rings with bright centre? 5
 (c) In a Newton's ring experiment, if air is replaced by a liquid of refractive index of 1.31, in what proportion would the diameters of the ring change? 4
- Q.7. (a) Define resolving power. Distinguish between prism and grating spectra. 3
 (b) Illustrate Fraunhofer diffraction at double slit and show that the angular separation between any two consecutive maxima is directly proportional to the wavelength of light used. 6
 (c) How many orders will be visible if the wavelength of incident radiation be 500 nm and the number of lines on the grating be 14000 an inch. 3
- Q.8. (a) State and explain Brewster's law. Discuss some of the applications of this law. 5
 (b) What do you understand by positive and negative crystals? Show that if an unpolarized light beam is incident on a polarizer the transmitted intensity will be the half the incident intensity. 5
 (c) Find the least thickness of a quarter wave plate which would convert plane polarized light into circularly polarized light. Given $\mu_o = 1.658$, $\mu_e = 1.486$ and wavelength of light is 5890 Å. 2

N.B. Answer **six** questions, taking **three** from each section.
The questions are of equal value.
Use separate answer script for each section.

SECTION-A

- ~~Q1.~~ (a) With reference to crystals, explain the terms (i) unit cell, (ii) single crystal and coordination member. What are the values of coordination members of sc, bcc and fcc structures? 3
- (b) Calculate packing fraction in case of bcc and fcc structures. $3\frac{1}{2}$
- (c) What are the parameters used to specify a unit cell? Give the unit cell specifications for cubic, orthorhombic and triclinic structures. 3
- (d) Distinguish between metal and insulator in terms of energy bands. $2\frac{1}{2}$
- Q2. (a) Show that the interference of two light waves causes the redistribution of light energy and the resultant interference pattern is in accord with the law of conservation of energy. 5
- (b) Show that the fringe width is given by $\beta = \frac{D\lambda}{d}$, symbols have their usual meanings. 4
- (c) Mention few practical applications of the interference from thin films. 3
- ~~Q3.~~ (a) Discuss Fraunhofer diffraction of monochromatic light from a single slit and show that the intensity of the secondary maximum is roughly 4.5% of that of the principal maxima. 6
- (b) Find the missing orders in a double slit diffraction pattern. 3
- (c) How many orders will be visible if the wavelength of incident radiation be $5000 \text{ \AA}$ and the number of lines on the grating be 14000 per inch? (1 inch = 2.54 cm) $\frac{\lambda}{d} = n\lambda$ 3
- ~~Q4.~~ (a) Explain plane of polarisation and plane of vibration for the plane polarised light. 3
- (b) Explain the action of (i) pile of plates and (ii) a tourmaline crystal in producing plane polarised light. 4
- (c) How a Nicol prism is used to analyse plane polarised light. 3
- (d) A 20cm long tube containing sugar solution is placed between crossed Nicols and illuminated by light of wave-length of $6000 \text{ \AA}$. If the specific rotation is 60° and optical rotation is 12° , what is the strength of the solution? $\frac{1}{2}$

SECTION-B

- Q5. (a) Based on Bohr's assumption, obtain the expression for the orbital energy of an electron in the hydrogen atom. 5
- (b) Explain the laws of photoelectric emission. Discuss the failure of classical theory to explain the experimental facts observed in photoelectric effect. 3
- (c) Describe the working of a photo-voltaic cell. 4
- Q6. (a) Obtain an expression for the change in wavelength of a photon undergoing Compton scattering. What is Compton wavelength? 6
- (b) Explain the reason of the presence of both modified and unmodified in the spectrum of Compton scattered radiation. 3
- (c) Show that the de Broglie phase velocity $v_p = f\lambda$ is always greater than the velocity of light in free space. 3
- ~~Q7.~~ (a) Define simple harmonic motion. Derive a general differential equation of motion of a simple harmonic oscillator and obtain its solution. 6

P.T.O

Q6) What are Lissajous figures? Calculate the resultant of two simple harmonic vibrations at right angle when their periods are in the ratio of 2:1 and there is a phase difference 0 and $\frac{\pi}{2}$.

Q8. (a) Explain free vibrations, forced vibrations and resonance by giving one example of each.

(b) For a plane progressive wave, show that on the average half the energy is kinetic and half potential.

(c) A wave of frequency 400Hz is travelling with a velocity 800ms^{-1} . How far are two points situated whose displacements differ in phase by $\frac{\pi}{4}$.

$$\frac{2\pi n}{\lambda}$$

$$\sqrt{=v}$$

$$k_2 \propto \omega^2$$

$$KE =$$